

Aazhmeer Chhapra

Artificial Intelligence Engineer

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OBJECTIVE

Artificial intelligence enthusiast with strong academic training and hands-on project experience in deep learning, time-series forecasting, computer vision, and generative AI. Seeking to apply data-driven modelling and intelligent systems to develop reliable, interpretable AI solutions for real-world applications, with a particular interest in healthcare and human-centred domains. Passionate about responsible AI development that balances technical performance with ethical and practical considerations.

Learnings & Skills:

- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Keras, LSTM, GRU, Transformers, Attention Models, CNNs, Transfer Learning, Hyperparameter Tuning
- **Time Series Forecasting:** Sequential Modeling, Financial Data Prediction, Technical Indicator Engineering
- **Natural Language Processing:** BERT, LLaMA2, LangChain, RAG, Sentiment Analysis, NER, Text Summarization
- **Computer Vision:** OpenCV, Image Classification, Object Detection, Feature Extraction
- **Programming Languages:** Python, SQL, JavaScript
- **Data Engineering & Analytics:** Data Pipelines, Preprocessing, Feature Engineering, Statistical Modeling, Matplotlib, Seaborn, Scikit-learn
- **Cloud & MLOps:** AWS SageMaker, Lambda, EC2, S3, Model Deployment, Distributed Training
- **Databases & Frontend:** PostgreSQL, Pinecone, ChromaDB, React.js, Vite
- **Frameworks & Tools:** Flask, Django, Django REST Framework, Git, GitHub, VSCode, Jupyter
- **Soft Skills:** Collaboration, Problem-Solving, Analytical Thinking, Communication, Teamwork, Adaptability, Fast Learner

EDUCATION & CERTIFICATIONS

MSc Artificial Intelligence — September 2026

Anglia Ruskin University, Cambridge, UK — Current

Modules: Advanced Machine Learning, Semantic Technologies, Applied Machine

Learning, Neural Computing and Deep Learning, Research Methods, Preparation for Professional Experience

Result: Distinction

Bachelor of Software Engineering — June 2024

COMSATS University Islamabad, Lahore

CGPA: 3.35

Dissertation: VirtuHire – An NLP-Powered Recruitment Platform using BERT and PyTorch

Certifications:

Data Science Bootcamp — 365 Data Science

AWS Academy ML Foundations — AWS Academy

Stanford Online — CS229, CS230

Projects

AstraLung: End-to-End Machine Learning Pipeline for Early Asthma Risk Prediction Aug 2025 - Present

- Integrated multi-year NHANES datasets (Demographics, Examination, Laboratory, Questionnaire) using pandas merge/join pipelines with SEQN-based respondent alignment, creating a unified clinical dataset of ~2K evaluated patient records for asthma-risk modelling.
- Performed extensive preprocessing including missing-value imputation, outlier filtering, One-Hot Encoding, and StandardScaler normalization, transforming heterogeneous clinical variables into a clean modelling feature space.
- Engineered clinically relevant features from pulmonary indicators, BMI categories, allergy markers, lifestyle factors, and blood biomarkers, improving downstream classifier performance during experimentation.
- Addressed class imbalance using SMOTE, increasing minority-class asthma recall to ~0.93, significantly improving detection of high-risk patients.

- Trained and compared Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and ensemble models, applying GridSearchCV hyperparameter tuning, with the final ensemble achieving 97.1% accuracy and F1-score 0.76 for asthma prediction.
- Conducted comprehensive EDA and statistical analysis (correlation matrices, distribution plots, chi-square tests, and feature importance analysis), identifying key predictors including BMI, allergy markers, and lifestyle variables associated with asthma risk.
- Documented reproducible pipelines and experimental results in a research-grade notebook, preparing the project for academic dissemination at a Cambridge-affiliated research venue.

End to End Medical Chatbot using GenerativeAI

Feb 2025 - Mar 2025

- Designed and implemented an NLP based AI-driven Medical Chatbot using LLaMA2 (LLM), integrating retrieval-based question answering with LangChain for accurate medical responses.
- Developed a Flask-based API to serve chatbot responses efficiently, ensuring seamless interaction between the user and the AI model.
- Utilized Pinecone vector database for fast and accurate semantic search, enhancing response relevance through retrieval-augmented generation (RAG).
- Optimized model inference with CTransformers, enabling lightweight execution of LLaMA 2 on local hardware.
- Engineered a user-friendly chat interface with HTML, CSS, and Flask, ensuring smooth and interactive communication.

Bird Sound Classification using CNN and Transfer Learning

Feb 2025 - May 2025

- Developed a pipeline to extract bird vocalizations from 24/7 unlabelled forest audio recordings provided by Cornell University using Librosa.
- Conducted comparative feature extraction using traditional CNNs, EfficientNetB0 (transfer learning), and AST (Audio Spectrogram Transformer).
- Evaluated model performance on clustering and classification tasks to identify the most effective feature extraction method.
- Visualized spectrograms and used MFCCs and log-mel features to enrich the representation of bird calls.
- Aimed to assist bioacoustic researchers in automatically identifying bird species from raw environmental recordings.

EXPERIENCE

Machine Learning Data Associate - II - Amazon AGI-DS Cambridge, UK

September 2025 – Present

- Performed high-volume data annotation for Amazon Nova by evaluating and ranking model-generated responses, consistently maintaining 100% quality scores while delivering precise comparative judgements aligned with internal evaluation standards.
- Worked across 15+ annotation workflows spanning diverse AI evaluation tasks, collaborating with cross-functional quality and workflow teams to clarify guidelines, resolve ambiguous cases, and ensure consistent labelling standards at scale.
- Achieved top-25% Average Handling Time (AHT) across the team while maintaining full annotation accuracy, and set a team record for the highest number of tasks processed in a single day, demonstrating exceptional productivity.
- Contributed to workflow improvement by identifying edge cases, reporting prompt and evaluation inconsistencies, and providing structured feedback that improved task clarity and operational efficiency across ongoing AGI data-labelling pipelines.

Junior Automation Engineer (Remote) – Pearl Control Systems, UK

November 2023 – May 2024

- Wrote and tested automation scripts for PLC, SCADA, and HMI systems for industrial control processes.
- Collaborated with software and engineering teams to integrate control systems with UI interfaces.
- Conducted end-to-end system testing with external clients from McCormicks, S4 Engineering, and Prologiks, ensuring high reliability and client satisfaction.
- Applied software development best practices to improve reliability, user experience, and documentation of automated workflows.